ZK-SNARK vs ZK-SCHNORR
Algorithm Comparison

Note: This comparison focuses on ALGORITHM CHARACTERISTICS ONLY. Network-dependent metrics (transfer time, throughput) are excluded.

Generated: 2025-11-30 16:05:09

1. FILE SIZE & STRUCTURE COMPARISON

Metric	zkSNARK (Groth16)	zkSchnorr	Analysis
Stego Image Size	507.78 KB (519,967 bytes)	506.72 KB (518,883 bytes)	zkSchnorr 1.06 KB smaller (0.2% difference)
Proof Size (embedded)	~192 bytes (3 G1 + 1 G2 points)	~300 bytes (signature + public key)	zkSNARK 108 bytes smaller (36% smaller proof)
Base Image	Same PNG	Same PNG	Identical source
Embedding Method	Chaos-based LSB	Chaos-based LSB	Identical method
Total Overhead	192 bytes proof + metadata	300 bytes proof + metadata	Both minimal overhead
File Format	PNG (lossless)	PNG (lossless)	Same format

2. NETWORK EFFICIENCY (Theoretical - Algorithm Impact Only)

Metric	zkSNARK	zkSchnorr	Explanation

TCP Segments Needed	357 segments (theoretical)	356 segments (theoretical)	1,084 bytes ÷ 1,460 MSS = 1 segment difference
Captured Segments	342 segments	341 segments	Actual Wireshark data (with optimizations)
Total Packets	369 packets	367 packets	2 packets difference (0.5%)
Control Overhead	27 packets	26 packets	1 control packet difference
Last Segment Size	397 bytes	773 bytes	zkSchnorr fills last segment better
Network Efficiency	**99.8% equivalent**	**99.8% equivalent**	Minimal difference in practice

3. CRYPTOGRAPHIC PROPERTIES

Property	zkSNARK (Groth16)	zkSchnorr
Proof Type	Zero-Knowledge Succinct Non-Interactive Argument of Knowledge	Schnorr Signature with Fiat-Shamir Transform
Cryptographic Basis	Elliptic Curve Pairing (BN254 curve, 254-bit)	Discrete Logarithm Problem (255-bit prime: 2^255-19)
Security Level	128-bit security	128-bit security
Zero-Knowledge	Full zero-knowledge proof (prover knows secret)	Signature-based proof (knowledge of private key)
Non-Interactive	Yes (after trusted setup)	Yes (via Fiat-Shamir)
Succinctness	Very succinct (~192 bytes) Constant size	Moderately succinct (~300 bytes) Linear in parameters

Verification Time	Very fast (~5-10ms) Constant time	Fast (~1-2ms) Linear time
Setup Phase	Requires trusted setup Powers of Tau ceremony	No trusted setup Deterministic generation

4. IMPLEMENTATION & DEPENDENCIES

Aspect	zkSNARK (Groth16)	zkSchnorr
Primary Language	Node.js + JavaScript	Python only
Circuit Compiler	Circom (domain-specific language)	Not required (direct implementation)
Proof Generation	snarkjs library + WASM witness calculator	Pure Python (hashlib, random)
Verification	snarkjs + verification key	Pure Python (standalone)
Key Management	Verification key required (~400 bytes JSON file)	Public parameters (in code)
Dependencies	Node.js, npm, snarkjs Circom compiler circomlib	Python 3.x only No external libraries
Installation Size	~200 MB (Node modules)	~0 bytes (standard library)
Deployment	Requires Node.js runtime Multiple files	Single Python file Self-contained
Portability	Moderate (platform-specific builds)	High (pure Python)

5. ALGORITHM COMPARISON CONCLUSIONS

#	Conclusion	Evidence & Explanation
1	**File Size is Nearly Identical**	zkSchnorr: 506.72 KB | zkSNARK: 507.78 KB Difference: 1.06 KB (0.2%) - NEGLIGIBLE for practical purposes. Both produce similar-sized stego images.
2	**Proof Size Paradox**	zkSNARK proof: ~192 bytes (smaller) | zkSchnorr proof: ~300 bytes (larger) BUT total file: zkSNARK larger by 1KB! Reason: PNG compression handles embedded data differently.
3	**Network Efficiency is Equivalent**	342 vs 341 TCP segments for ~507 KB transfer. 1 segment difference (0.3%) is algorithmically insignificant. Both systems have equivalent network footprint.
4	**zkSchnorr: Simpler Dependencies**	Python-only, zero external dependencies, self-contained. Easier deployment, smaller installation, higher portability. Ideal for resource-constrained or isolated environments.
5	**zkSNARK: Stronger Cryptography**	Full zero-knowledge proof with succinctness properties. Smaller proof size, faster verification (pairing-based). Trade-off: Requires trusted setup and more dependencies.
6	**Both Systems Are Secure**	Both provide 128-bit security level. zkSNARK: Pairing-based assumptions | zkSchnorr: Discrete log. Security is equivalent in practice.

IMPORTANT: Transfer speed (13.6s vs 0.1s) is NOT an algorithm characteristic. It depends on network conditions (bandwidth, latency, congestion) at the time of measurement. Both algorithms produce similar file sizes (~507 KB) and require similar network resources (~341-342 packets). The 133x speed difference observed was due to network conditions, not algorithm design.